

Enterprise AI Workflow Evaluation Framework

How We Benchmark, Validate, and Guarantee Production-Grade Automations

TARGET: Enterprise Operations & Tech Leadership
SCOPE: Automated Workflow Reliability (Evals)
STATUS: **Production Hardening Standard**

The HashCase Guarantee: General-purpose LLMs are inherently probabilistic, but enterprise operations require 100% deterministic guarantees. At HashCase, we do not deploy raw prompts into your organization. We engineer, stress-test, and deploy hardened AI workflows governed by a 6-pillar automated evaluation framework. Every automated action is validated for business logic correctness, API precision, cost predictability, and security before touching client systems.

Our 6 Core Evaluation Pillars

PILLAR 1: Deterministic & Programmatic Business Logic Evals

Eliminating Silent Failure through Code-Level Assertions

What We Do: We do not rely on AI to verify its own factual or arithmetic correctness. Before any workflow output or database update is approved, our automated harness runs strict code-level assertions: validating strict data schemas (JSON/Pydantic), confirming relational invariants (e.g., invoice totals = subtotal + tax), and verifying foreign keys and entity IDs directly against your enterprise records.

Client Guarantee: Zero hallucinated schema fields, 100% mathematical consistency, and guaranteed type safety.

PILLAR 2: Multi-Step & Agent Trajectory Evals

Tool Selection Accuracy, Trajectory Efficiency & Loop Detection

What We Do: Modern automations chain multiple tools (CRMs, ERPs, APIs, databases) together. We evaluate the entire execution path: measuring tool selection precision/recall (ensuring the agent selects the exact right tool with correct arguments), tracking trajectory efficiency (eliminating wasteful intermediate calls), and enforcing deterministic loop detection to ensure an agent never gets trapped in infinite execution cycles.

Client Guarantee: Predictable, streamlined agent execution without runaway API loops, wasted tokens, or misfired webhooks.

PILLAR 3: LLM-as-a-Judge with Calibrated Rubrics

Automated Qualitative Audits Calibrated to Your Senior Leadership

What We Do: For qualitative outputs (customer emails, policy summaries, case triage), we deploy frontier LLM judges governed by exhaustive, multi-point rubrics (evaluating faithfulness to source documents, tone, completeness, and policy compliance). Crucially, we calibrate our automated judges against your human Subject Matter Experts (SMEs) to ensure our automated evaluations achieve >90% statistical alignment with your senior staff.

Client Guarantee: Automated, scalable quality control that grades every output with the exact standards of your best employees.

PILLAR 4: Sandbox Simulation & Mock Environments

Safe, Side-Effect-Free Validation Against Enterprise Systems

What We Do: Testing live AI agents against real production databases or ticketing platforms creates immense operational risk. HashCase evaluates workflows in isolated sandboxes and mock API environments. We validate state transitions—confirming not just what the model outputs, but that the mock CRM record, inventory count, or status flag transitioned to the exact intended state under simulated network delays, rate limits, and service outages.

Client Guarantee: Zero risk to production data, with proven state-level verification before live deployment.

PILLAR 5: Curated 'Golden Datasets' & Synthetic Expansion

Continuous Stress-Testing Against Adversarial Edge Cases

What We Do: We curate a proprietary regression test suite from your historical enterprise records and real-world edge cases. We then artificially expand this dataset with synthetic adversarial generation: intentionally injecting typos, missing fields, contradictory customer requests, and malformed inputs. Crucially, every production anomaly or corner-case discovered is converted into a permanent regression test case.

Client Guarantee: *An automation system that becomes systematically more resilient with every iteration, preventing regression.*

PILLAR 6: Operational, Cost & Security Evals

Latency Budgets, Token Attribution & Adversarial Jailbreak Defense

What We Do: Enterprise readiness requires meeting strict operational SLAs. We benchmark Time-to-First-Token (TTFT) and P95/P99 execution latency. We enforce granular cost-per-workflow attribution to ensure high ROI and intelligent model routing. Furthermore, we execute automated security red-teaming to defend against indirect prompt injection, jailbreaks, and unauthorized PII leakage.

Client Guarantee: *Predictable cloud unit economics, guaranteed SLA compliance, and enterprise-grade data privacy.*

HashCase Evaluation Architecture at a Glance

Evaluation Layer	Primary Metrics & Checks	HashCase Safeguard for Your Business
Deterministic & Logic	Schema validation, arithmetic consistency, foreign key matching	Zero corrupted database records or malformed payloads
Agent Trajectory	Tool selection accuracy, step-count efficiency, loop breaker	Eliminates runaway API costs and infinite retry cycles
LLM-as-a-Judge	Groundedness/faithfulness, tone, policy compliance, SME rubrics	Continuous auditing matching senior leadership's standards
Sandbox Simulation	Mock CRM/ERP API calls, state transitions, failure recovery	Guarantees safe execution with zero production risk
Golden Datasets	SME historical baselines, synthetic adversarial stress cases	Prevents regressions; automations strengthen over time
Operational & Security	P95/P99 latency, TTFT, cost-per-run, prompt injection defense	Ensures strict SLA compliance, budget control & data privacy

Continuous Production Monitoring: Beyond Deployment

Our evaluations do not end at deployment. HashCase workflows operate with Shadow Deployments (auditing against human staff), Human-in-the-Loop confidence thresholds (routing borderline decisions to supervisors), and continuous drift monitors that alert our engineers the moment upstream LLM model updates or data distributions shift.

Deploy AI with Deterministic Confidence. Contact HashCase to schedule an AI Evaluation Audit or discuss hardening your automated workflows: hello@hashcase.com | www.hashcase.com